DistilSCD: Output Distillation of a Remote-Sensing Foundation Model into a Compact Semantic Change Detector

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***Abstract*—Remote-sensing foundation models hold the best published accuracy for semantic change detection (SCD), at a cost that is hard to deploy at scale: the strongest SECOND model, PerASCD, has 548.17 M parameters and needs about 1.5 TMACs per image pair. We ask how much of that accuracy a compact model inherits from the teacher's outputs alone. DistilSCD trains a 16.48 M-parameter siamese VMamba student with two extra loss terms, soft change targets and temperature-scaled class distributions, read from an offline cache of the teacher's logits under the eight D4 views that the student's augmentation produces; inference cost is unchanged. Hyper-parameters were selected on an internal validation split. On SECOND (five seeds), distillation raises the student from 23.90 to 26.20 SeK (+2.30 [2.13, 2.47]), equivalent to the teacher within ± 0.25 SeK, with 33.3$\times$ fewer parameters, 51.2$\times$ fewer MACs and 4.9–12.5$\times$ lower GPU latency. Because the released random pair split of Landsat-SCD places both images of every test pair in the training set, we introduce a location-disjoint four-fold protocol; under it, distillation improves the student on every fold and seed (+2.53 SeK on the three primary folds). Controls with identical losses and cache tie the gain to the teacher's targets: on these folds, neither self-distillation nor ground-truth targets smoothed to the teacher's confidence improve the student, and the foundation-model targets exceed them by +2.61 and +2.75 SeK; on SECOND, self-distillation recovers about one third of the gain, and a decaying learning rate does not improve the undistilled baseline.**

***Index Terms*—Semantic change detection, knowledge distillation, foundation models, state-space models, efficient deep learning, evaluation protocol, spatial cross-validation, Landsat.**

I. Introduction

SEMANTIC change detection (SCD) maps *where* the land surface changed between two acquisition dates and *what* it changed from and to. It underlies land-use monitoring, urban-growth accounting and disaster assessment, where the same model is run over very large archives of image pairs. Accuracy on the standard benchmarks has risen steadily, from multitask convolutional networks [1], [2] through bi-temporal semantic reasoning and spatio-temporal Transformers [3], [4] to state-space models [5], [6]. The current best results on SECOND come from remote-sensing foundation models fine-tuned for the task: PerASCD [7] adapts a 40-block PerA ViT-G [8] with a ViT-Adapter [9] and reaches 26.11 SeK. That accuracy comes with 548.17 M parameters, about 1.5 TMACs per pair, and more than 25 GB of GPU memory at batch size eight in our measurements, an order of magnitude beyond what efficient SCD models use [5].

Knowledge distillation (KD) [10] offers a way to keep the accuracy and drop the cost (Fig. 1). Distilling foundation models into efficient change detectors has been explored for *binary* change detection, either through features [11] or through output probabilities [12]. Whether it works for *semantic* change detection, how much of the teacher's accuracy a compact student can reach, and *why* it works, remain open. The KD literature predicts both outcomes: very large teachers can be hard to imitate for small students [13], [14], yet students sometimes exceed their teachers [15]–[17], and part of the effect of KD can be explained as label smoothing rather than transferred knowledge [16], [18].

Answering these questions requires an evaluation that separates the effect of the teacher from seed noise, from model selection on the test set, and from the benchmark itself. The latter turned out to matter. Landsat-SCD [19], the second standard SCD benchmark, is built from a small number of tiles observed at many dates. We show that a random split at the level of image pairs, the dataset's original protocol, places the same images in the training and test sets, so that the test set measures recognition at locations and dates already seen in training rather than generalisation to new locations. Random splits of spatially structured data are known to answer this narrower question [20]–[23], and it changes what a distillation experiment on this dataset can conclude.

This paper makes the following contributions.

1) *DistilSCD, an output-only distillation recipe for SCD.* A frozen foundation-model teacher is run once, offline, under the eight D4 views that the student's augmentation produces; its change logits and class distributions supervise the student through a soft binary cross-entropy and a temperature-scaled KL term. The student's architecture and inference cost are unchanged, and training takes the same time as without distillation (Section III).
2) *A controlled evaluation on SECOND.* With distillation hyper-parameters selected on an internal validation split and five seeds on the test set, a 16.48 M-parameter student gains 2.30 SeK and is equivalent to the 548.17 M-parameter teacher within ± 0.25 SeK at 51.2$\times$ fewer MACs. A second, smaller and ImageNet-pretrained teacher yields a smaller gain, and its student

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Computations were performed on the Narval cluster of the Digital Research Alliance of Canada.

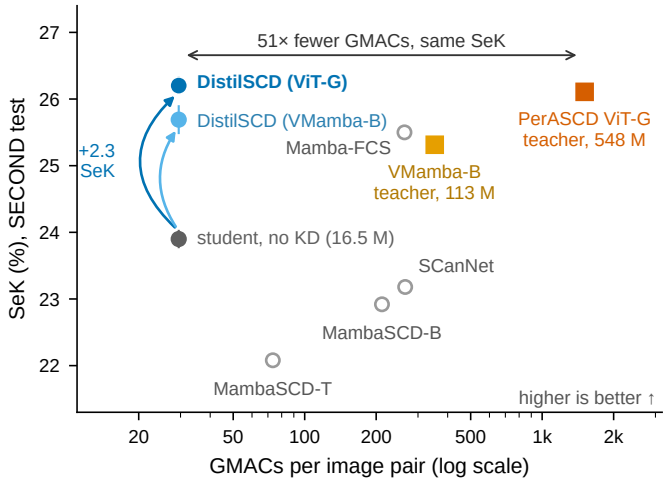


Fig. 1. Accuracy versus inference cost on the SECOND test set (official split). Distillation moves the 16.48 M-parameter student from 23.90 to 26.20 SeK with the ViT-G teacher and to 25.69 with the VMamba-B teacher, at unchanged cost. Students: mean over seeds of the best test epoch (field convention; last-epoch values in Table IV); teachers: published checkpoints re-evaluated by us. Our models and the teachers are counted with the same counter (PyTorch aten), which gives the 51.2× ratio; published models (open circles) are placed at their published complexity, an fvcore-style MAC count, so the two groups of counts are not strictly comparable. SCanNet: as retrained on the official split by [7].

exceeds it (Section V).

3) *A train–test overlap analysis and a location-disjoint protocol for Landsat-SCD.* We verify at the pixel level that the dataset holds 333 distinct images recombined into 2 385 pairs, and release a four-fold cross-validation in which test tiles are never seen in training (Section IV-B).
4) *Controls that locate the gain in the teacher's targets.* Under the location-disjoint protocol the distilled student improves on its undistilled counterpart on every fold and seed and exceeds the single-run fold teachers. With the same losses, cache and schedule, replacing the foundation-model targets by the undistilled student's own outputs (self-distillation) or by ground truth smoothed to the teacher's confidence removes the gain on these folds and leaves about one third of it on SECOND; a decaying learning rate does not improve the undistilled baseline, and our feature-distillation term adds nothing on top of output distillation (Sections VI–VIII).

Code, splits, the overlap-verification script and the run logs are released (see Data and Code Availability).

II. Related Work

A. Semantic Change Detection

HRSCD [1] framed SCD as joint land-cover mapping and change detection. Yang *et al.* [2] introduced the SECOND dataset, an asymmetric siamese network and the separated kappa (SeK) metric used throughout this paper. Bi-SRNet [3] and SCanNet [4] added bi-temporal semantic reasoning and spatio-temporal Transformer modelling, and ChangeMask [24] decoupled semantic and change branches. State-space models entered the field with ChangeMamba [5], built on Mamba [25] and VMamba [26], and were followed by Mamba-FCS [6] and GSTM-SCD [27]. The strongest recent results use foundation-model encoders: PerASCD [7], DBTANet [28] with a SAM branch, and SemDINO [29]. Lightweight change detection has mostly been pursued by architecture design for binary change [30]–[32]; Corley *et al.* [33] caution that well-trained simple baselines remain competitive. We propose no new architecture: our contribution is a training signal and an evaluation protocol.

B. Foundation Models for Remote Sensing and Change Detection

Remote-sensing foundation models are pre-trained by masked image modelling or contrastive objectives on large archives [8], [34]–[45]. For change detection they are usually kept inside the deployed network, either fine-tuned [7], [28], [46]–[49] or used zero-shot [50]. Benchmarking studies report that such comparisons are sensitive to evaluation details [51], [52]; we encountered the same sensitivity when reproducing a published teacher (Section IV-F). Here the foundation model is used only at training time.

C. Knowledge Distillation

Output distillation [10] and its feature-based variants [53]–[57] are surveyed in [58]. Two lines of results bear on our setting. First, the *capacity gap*: larger teachers do not always make better teachers for small students [13], [14], [59], although Beyer *et al.* [60] show that consistent teaching (teacher and student see the same view) and long schedules close much of the gap. Our D4 cache makes teaching geometrically consistent, akin to stored per-crop soft labels [61], [62]; photometric jitter is applied to the student only, so teaching is not fully consistent in their sense. Second, students can outperform their teachers [15], [16], [63], and students that generalise well may match their teachers poorly [17]; Yuan *et al.* [16] interpret part of KD as a learned label smoothing [18], [64]. For dense prediction, structured and channel-wise distillation transfer spatial structure for segmentation [65]–[68]. Closest to our recipe in general vision, Vemulapalli *et al.* [69] fine-tune a vision foundation model on the target task before distilling it into a small task model.

D. Distillation in Remote Sensing and Change Detection

KD in remote sensing is surveyed in [70]. For change detection, CDKD [12] distils a large teacher's output probabilities, with contrastive and normalised terms, into a lightweight on-board binary detector, and several works align features between large and small binary change detectors [71]–[73]. Burden-free distillation (BFD) [11] is the closest in aim: it transfers features of generic vision foundation models into binary change detectors and removes the foundation model at inference. Other teacher–student work targets accuracy or robustness rather than compression: multi-teacher distillation [74], self-distillation [75]–[77], weak supervision [78], and GUIDE [79], a teacher–student framework for SCD under

cross-temporal domain shift. Relative to this work, we study *semantic* change detection on SECOND and Landsat-SCD, distil only the *outputs* of a task-fine-tuned remote-sensing foundation model, and test where the gain comes from with ground-truth target controls, two teachers and location-disjoint folds.

E. Evaluation Leakage and Spatial Validation

Random splits of spatially or temporally dependent samples inflate accuracy; blocked or spatial cross-validation is the remedy [20], [21], [80], [81]. In remote sensing, spatial dependence between training and test samples is a documented pitfall of accuracy assessment [22], [82]–[84]. Kapoor and Narayanan [23] classify non-independence between training and test samples as leakage unless the scientific claim concerns a distribution with the same dependence structure. That qualification frames our treatment of Landsat-SCD: its random split is a valid protocol for change recognition over one already-seen area, and we change the question to generalisation to unseen locations. Reusing a test set for model selection biases reported accuracy [85], [86], which motivates reporting a selection-free number next to the field's convention, and seed variance motivates paired multi-seed comparisons [87].

III. Method

Fig. 2 summarises DistilSCD.

A. Task and Notation

An input is a co-registered image pair (x^1, x^2) of size $H \times W$. The labels are a binary change map $y^{\mathrm{c}} \in \{0,1\}^{H\times W}$ and two semantic maps $y^1, y^2 \in \{0,\dots,C-1\}^{H\times W}$, where class 0 denotes "no change" (no semantic label) and classes $1,\dots,C-1$ are land-cover classes, defined only on changed pixels. SECOND has $C = 7$ (six land-cover classes); Landsat-SCD has $C = 5$ (four classes).

B. Student

The student is a compact siamese network developed by the first author before this study and kept frozen in design and recipe throughout. A shared VMamba encoder [26] with depths $(2,2,5,2)$, widths $(96,192,384,768)$ and the MLP sub-blocks disabled ("mini", 13.84 M parameters) processes each date and is initialised from ImageNet VMamba-T weights. Features of the two dates are fused per stage by concatenation and a 1×1 projection; a binary decoder predicts change logits $b \in \mathbb{R}^{2\times H\times W}$, and a semantic decoder shared between the dates, conditioned on a learned temporal embedding and on the change maps of the binary decoder, predicts class logits $s^1, s^2 \in \mathbb{R}^{C\times H\times W}$. Both decoders use depth-wise refinement and DySample upsampling [88]. The model has 16.48 M parameters and 29.48 GMACs per 512×512 pair (31.42 with fvcore). Two blocks inherited from the student's earlier design (a frequency branch and a change-guided attention) have no established effect on SECOND; they are kept unchanged and documented in the supplementary material, together with the design experiments, which selected epochs on the SECOND test set as is customary on this dataset.

The task loss is

$$\begin{aligned}\mathcal{L}_{\mathrm{task}} = \mathrm{CE}(b, y^{\mathrm{c}}) + \lambda_{\mathrm{sc}}\,\mathcal{L}_{\mathrm{sc}} \\ + \tfrac{1}{2}\big[\mathrm{CE}_{\neq 0}(s^1, y^1) + \mathrm{CE}_{\neq 0}(s^2, y^2)\big],\end{aligned} \tag{1}$$

where $\mathrm{CE}_{\neq 0}$ ignores class-0 pixels and $\mathcal{L}_{\mathrm{sc}}$ ($\lambda_{\mathrm{sc}} = 0.1$) is the bi-temporal feature-consistency loss of Bi-SRNet [3].

C. Teachers

The main teacher is the published SECOND checkpoint of PerASCD [7]: a PerA ViT-G/16 encoder (width 1024, 40 blocks) [8], a ViT-Adapter [9] and a cascade gated decoder, 548.17 M parameters in total. It outputs, at $1/4$ resolution, class logits $z^1, z^2 \in \mathbb{R}^{C\times h\times w}$ for each date and a change logit $z^{\mathrm{c}} \in \mathbb{R}^{h\times w}$, which it upsamples bilinearly to full resolution. The second teacher, released by the same authors, replaces the encoder with VMamba-B (113.01 M parameters). Both are frozen.

D. Offline, View-Consistent Logit Cache

The student is trained with horizontal and vertical flips and 90° rotations, so each training sample is seen under one of the eight elements g of the dihedral group D_4. The teacher is not equivariant to these transforms: on SECOND, its change decision for $T(g\cdot x)$ and $g\cdot T(x)$ differs on 2.2–2.5 % of all pixels (about 20 % of the pixels are changed), and its class decision on 2.7–3.0 % of the changed pixels. We therefore run the teacher once on every training pair under all eight views and store the $2C+1$ output channels at their native $1/4$ resolution in half precision (about 12 GB for the 2 968 SECOND pairs, computed in 26 GPU-minutes). During training, the cache entry of the drawn view g supervises the augmented sample, so teacher and student always see the same geometry; when the two dates are swapped (temporal-swap augmentation), the two semantic blocks of the target are swapped and the symmetric change logit is kept. Photometric jitter is applied to the student's input only, which adds a radiometric invariance constraint and makes teaching photometrically inconsistent. We did not ablate the eight-view cache against a single view. The cached logits are upsampled bilinearly to the student's resolution, exactly as the teacher's own last operation.

E. Distillation Losses

Change distillation. With the student's change logit $d = b_1 - b_0$ and temperature T_{c},

$$\mathcal{L}_{\mathrm{chg}} = T_{\mathrm{c}}^2\,\mathrm{BCE}(\sigma(d/T_{\mathrm{c}}),\ \sigma(z^{\mathrm{c}}/T_{\mathrm{c}}))\,, \tag{2}$$

averaged over all pixels (the change map is supervised everywhere). We use $T_{\mathrm{c}} = 1$.

Semantic distillation. Class 0 is never a target of the teacher's cross-entropy, so its logit carries no supervised meaning. We renormalise both distributions over the real classes $1,\dots,C-1$:

$$\mathcal{L}_{\mathrm{sem}} = \frac{T_{\mathrm{s}}^2}{2}\sum_{t\in\{1,2\}}\frac{1}{|\mathcal{M}|}\sum_{u\in\mathcal{M}}\mathrm{KL}\big(q_u^t \,\|\, p_u^t\big)\,, \tag{3}$$

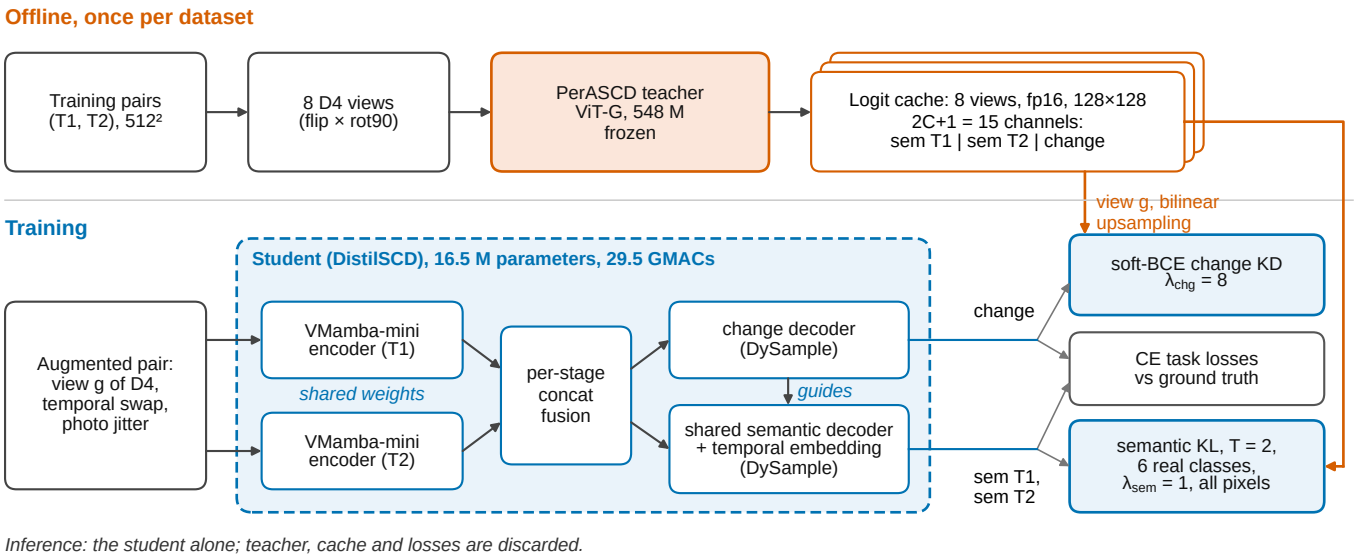


Fig. 2. DistilSCD. Offline and once per dataset, the frozen teacher is run on the eight D_4 views of every training pair; its $2C+1$ output channels (class logits of both dates, class 0 = no change, and a change logit) are stored at 128×128 in half precision. During training, the cache entry of the view drawn by the student's augmentation is upsampled bilinearly and supervises the student through a soft-target BCE on the change logit ($\lambda_{\mathrm{chg}} = 8$) and a temperature-scaled KL on the class distributions renormalised over the real classes ($T_{\mathrm{s}} = 2$, $\lambda_{\mathrm{sem}} = 1$, all pixels), in addition to the task losses. Only the student is used at inference. Blocks inherited from the student's earlier design with no measurable effect are omitted (supplementary material).

with $q^t = \mathrm{softmax}(z^t_{1:C-1}/T_{\mathrm{s}})$ and $p^t = \mathrm{softmax}(s^t_{1:C-1}/T_{\mathrm{s}})$. The mask $\mathcal{M}$ is either the whole image or the pixels predicted changed by the ground truth or the teacher; on SECOND, about 80 % of pixels have no semantic label, so the first option asks the student to follow the teacher's semantics where neither model was supervised.

Objective. $\mathcal{L} = \mathcal{L}_{\mathrm{task}} + \lambda_{\mathrm{chg}}\mathcal{L}_{\mathrm{chg}} + \lambda_{\mathrm{sem}}\mathcal{L}_{\mathrm{sem}}$. The retained configuration, selected as described in Section IV-E, is $\lambda_{\mathrm{chg}} = 8$, $\lambda_{\mathrm{sem}} = 1$, $T_{\mathrm{s}} = 2$ and $\mathcal{M}$ = all pixels. We also screened one feature-distillation term (cosine similarity after channel-wise normalisation between 1×1-projected student features and the teacher's encoder features at two or three scales) on top of output distillation; it requires running the teacher online.

F. Training Recipe

All students, with and without distillation, use the same recipe: full 512×512 tiles, micro-batch 2 with 4 accumulation steps (effective batch 8), AdamW (learning rate 10^{-4}, weight decay 0.01) with a 1 500-iteration linear warm-up and a constant rate afterwards, bfloat16 autocast, an exponential moving average of the weights (decay 0.9998) used for evaluation, flips, 90° rotations, photometric jitter (strength 0.2) and temporal swap (probability 0.5). The budget is 120 epochs on SECOND (44 520 optimiser steps) and 100 epochs on Landsat-SCD (Section IV-C). The constant rate comes from the student's design campaign; with it, a shorter run is exactly a prefix of a longer one; the seven no-distillation SECOND seeds were trained for 200 epochs and are read at epoch 120. Distillation adds no measurable student training time, since the teacher never runs during student training; the teacher's cost is paid once, when the cache is computed (and, on Landsat-SCD, when the fold teachers are trained).

IV. Datasets and Evaluation Protocol

Table I summarises the datasets, splits and model-selection rules.

A. SECOND

SECOND [2] contains 4 662 pairs of 512×512 aerial images with six land-cover classes, split officially into 2 968 training and 1 694 test pairs; there is no official validation split. Several papers evaluate on a split of the public training set instead of the official test set [3], [4]; we use the official split throughout and compare only with results obtained on it. A pixel-hash check found no exact duplicate image between the SECOND training and test sets.

B. Landsat-SCD and Its Structure

Landsat-SCD [19] covers a single area of Tumushuke, China, at 30 m resolution, with 28 acquisition dates between 1990 and 2020 and four land-cover classes (farmland, desert, building, water). We use the 512×512 version distributed with PerASCD, which removes the offline-augmented copies of the original release and keeps 2 385 pairs split at random into 1 431/477/477 for training, validation and test.

Pair names encode two years and a tile index k taking only 12 values. A name-independent, pixel-level verification (Table II) shows that the 2 385 pairs are built from 333 distinct images, i.e. 12 tiles observed at about 28 dates, and that each (tile, year) group contains exactly one image. As a consequence, all 477/477 test pairs, and all 477/477 validation pairs, have both of their images in the training set, with identical class labels on the 198.8 M annotated pixels they share. A learning-free lookup that copies, for each test pixel, the classes annotated for the same tile and year in the training

TABLE I
DATASETS AND EVALUATION PROTOCOLS (NUMBER OF IMAGE PAIRS).

Dataset	Protocol	Train	Val	Test	Selection / remark
SECOND	official split	2 968	–	1 694	max test SeK over epochs and final epoch
SECOND	internal validation (screening)	2 671	297	–	max validation SeK; test never used
Landsat-SCD	released random pair split	1 431	477	477	test pairs with both images in train: 477/477
Landsat-SCD	location-disjoint fold 0 (exploratory)	1 603	202	580	test tiles 00, 02, 20; val tile 21
Landsat-SCD	location-disjoint fold 1	1 590	202	593	test tiles 01, 21, 30; val tile 12
Landsat-SCD	location-disjoint fold 2	1 590	189	606	test tiles 10, 12, 22; val tile 02
Landsat-SCD	location-disjoint fold 3	1 577	202	606	test tiles 11, 31, 32; val tile 10

Landsat-SCD holds 333 distinct images at 12 locations, recombined into pairs. Location-disjoint folds: each location is a test location in exactly one fold; test at the epoch of best validation SeK of the fold. SECOND has no official validation split.

TABLE II
TRAIN–TEST IMAGE OVERLAP IN THE RELEASED LANDSAT-SCD RANDOM PAIR SPLIT (LANDSATSCD512).

Check	Value
Image slots (2 images × 2385 pairs)	4 770
Distinct images by pixel hash	333
Tile indices k (locations)	12
Val pairs with both images present in train	477 / 477
Test pairs with both images present in train	477 / 477
(k, year) groups with a single distinct image	333 / 333
Annotated pixels compared across duplicates	198 829 642
Class agreement on those pixels (%)	100.0
Lookup oracle SeK on test (no learning)	77.87
Lookup oracle Fscd on test	95.04
Lookup oracle mIoU on test	94.25
Test changed pixels whose two classes are known from train (%)	90.8
SECOND exact train/test duplicates (5936 train, 3388 test images)	0

From scripts/verify_landsat_leakage.py (pixel hashes of every image slot). The lookup oracle predicts each test pair by looking up the training labels of the same two images, without learning. Last row: same check on SECOND.

set reaches 77.87 SeK on the test set (Fscd 95.04, mIoU 94.25), higher than the published Landsat-SCD results we are aware of, which were obtained on random pair splits (e.g. 65.21 for PerASCD [7]). Each image appears in about 14 pairs; when 60 % of the pairs are drawn at random for training, the probability that a test image appears in none of the training pairs is about $0.4^{13} < 10^{-5}$, so any random pair split has this property, whatever its seed. We verified it on the released split only.

This is not an error of a particular paper. The original authors split pairs at random by design [19, p. 11], for a dataset built as a long time series of one area, and name the extension to several regions as future work [19, p. 19]. The random split measures change recognition on an area already seen at other dates; it does not measure generalisation to a new area. Our student illustrates the difference (Fig. 3b): on the random split its test SeK keeps rising, from 59.85 at 250 epochs to 69.42 at 600 epochs, with the validation curve still climbing by 0.18 points per 10 epochs at the end, whereas on unseen tiles it plateaus below 30 SeK within 100 epochs.

C. Location-Disjoint Cross-Validation on Landsat-SCD

We group pairs by tile and build four folds of three test tiles each, chosen by exhaustive search over the 15 400 partitions of the 12 tiles into four groups of three to balance the transition types and the change ratio (test change ratio 0.184–0.191, overall 0.188). In each fold, one of the nine remaining tiles serves as validation and eight as training; the training code refuses to start if a tile appears in two subsets. Test sets hold 580 to 606 pairs. The student's epoch budget was fixed once, on fold 0 and on validation only, as the first 50-epoch window in which the validation SeK slope fell below 0.05 points per 10 epochs (epochs 50–99), giving 100 epochs for every arm and fold. The folds, the verification script and the budget rule are released.

D. Metrics, Model Selection and Statistics

We report SeK [2], the SCD F-score Fscd and the mIoU of the change/no-change map, computed with one implementation that reproduces the published teacher scores to within 0.004 points. Values are in percentage points.

Model selection. On SECOND, which has no validation split, the field reports the best test SeK over epochs [5], [7]. We report this number and, next to it, the SeK at the last epoch, which involves no selection. On Landsat-SCD we report the test SeK at the epoch of best validation SeK of the fold.

Statistics. Each arm is trained with several seeds; a seed fixes initialisation, data order and augmentation draws, so arms with the same seed share their first mini-batch exactly. On SECOND we report Welch tests and 95 % confidence intervals for differences of means, paired tests across seeds, and one-sample tests against the single-seed teacher. On Landsat-SCD, the variation between folds (standard deviation 4.54 SeK between fold means of the undistilled student) dwarfs the variation between seeds, so all comparisons are paired by fold and seed, and the unit of inference is the fold: we report a t interval over the fold means. As a consistency check, not a test, we also count the fold–seed pairs with a positive difference (seeds within a fold are not independent). To compare the distilled student with the single-run teacher on SECOND we use an equivalence test (two one-sided tests, margin ± 0.25 SeK), which treats the teacher's score as fixed. GPU training is not bit-reproducible; replays of the same configuration differ by up to 0.5 SeK on Landsat-SCD, which adds variance to each paired difference.

E. Screening Specified in Advance

All distillation hyper-parameters were chosen on SECOND before any distillation run was evaluated on the test set. The

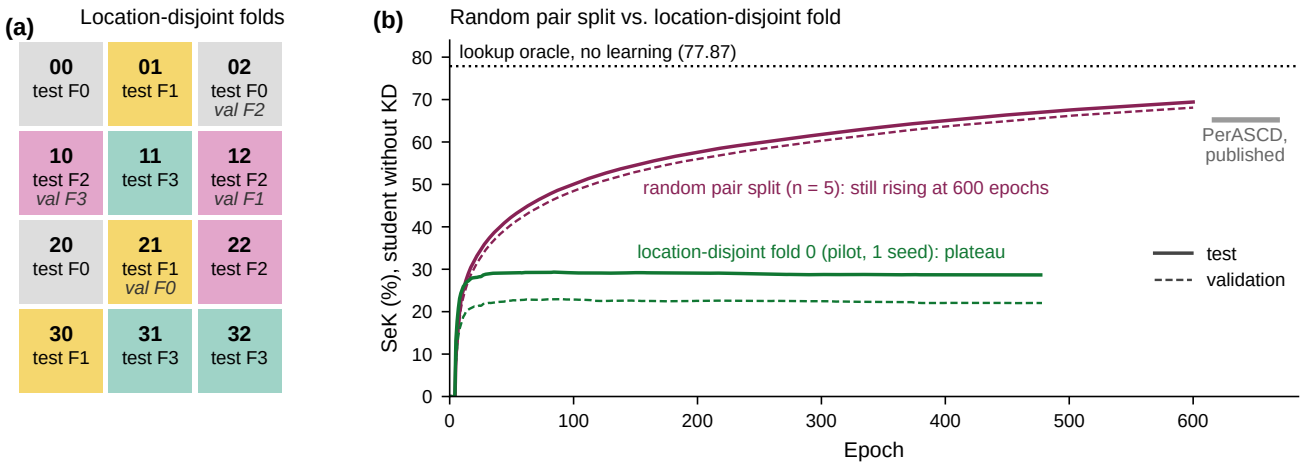


Fig. 3. Landsat-SCD: train–test image overlap of the released random pair split, and location-disjoint protocol. (a) The 12 tile indices k drawn as a 4×3 grid (row = first digit, column = second digit; the layout is inferred from the indices, not from georeferencing), coloured by the fold in which the tile is a test tile; "val Fi" marks the validation tile of fold i. Fold 0 is exploratory. (b) Test (solid) and validation (dashed) SeK per epoch of the same undistilled student on the released random split (mean of five seeds; 69.42 at epoch 600, still rising) and on location-disjoint fold 0 (one pilot seed, stopped at epoch 477). Dotted line: learning-free lookup oracle (77.87). Grey tick: PerASCD as published (65.21). The two sets of numbers answer different questions and are not comparable.

2 968 training pairs were split into 2 671 for training and 297 for validation (a fixed, hashed split). With two seeds per arm and 120 epochs, three waves fixed in turn the change weight λ_{chg}, the semantic term ($T_{\text{s}} \in \{2, 4\}$, two masks) and the feature term (two sets of scales, two weights). Before each wave, the decision rule ("best mean validation SeK; an increment counts if it exceeds 0.36 points") was written in the project's lab journal, which is versioned with the code; it is not a public registration. The 0.36-point threshold was derived for five test seeds and serves here as a ranking threshold, not as a test. The λ_{chg} grid started at $\{0.5, 1, 2\}$ and was extended to $\{4, 8\}$, then to 16, because the best value lay at the edge of the grid. Two controls, run at $\lambda_{\text{chg}} = 2$, replaced the teacher's change target by the ground truth, either binary or average-pooled to the teacher's resolution and upsampled like it. The SECOND teacher was trained by its authors on all 2 968 training pairs, including our 297 validation pairs, which can favour distillation over the controls on validation; Section IV-G repeats them at the retained configuration on held-out data. The retained configuration was then retrained on all 2 968 pairs with five seeds and evaluated on the test set with the field's protocol (Section IV-D).

F. Teachers: Reproduction and Retraining

Re-evaluated with our code, the published PerASCD checkpoint gives 26.1079 SeK on the SECOND test set (published: 26.11). The published VMamba-B checkpoint gives 23.73 with the PerA normalisation used by the released evaluation code, and exactly its stored score, 25.314, with ImageNet normalisation, which we therefore use. Both checkpoints were selected by their authors on the test set and are single runs; the training log of the ViT-G checkpoint reads 26.13 at its best epoch and 25.85 at its last epoch. No Landsat-SCD teacher trained on unseen tiles exists, and the published one has seen our test images; we therefore retrained one PerASCD ViT-G teacher per fold with the authors' recipe (50 epochs, SGD), selected on the fold's validation tile. Activation checkpointing of the 40 ViT blocks, which leaves gradients unchanged, fits training on a 40 GB A100. Before caching each fold teacher, a gate specified in advance required its validation SeK to exceed the mean validation SeK of the three undistilled students of that fold.

G. Control Arms

After an internal review of a first draft, we added three controls; their comparisons, selection rules and statistics were written in the lab journal before any of these runs was evaluated. All other settings are those of DistilSCD, including both loss terms and their weights. (i) *Self-distillation.* The teacher is replaced by the undistilled student of seed 1 (on SECOND its best test epoch, 23.87 SeK, the selection used for the published teacher; on Landsat-SCD its best validation epoch on each fold), cached under the same eight views and in the same format. Students use seeds 2–4 on SECOND and 2–3 on each fold, so that none shares its teacher's initialisation. The student predicts at full resolution; its logits are average-pooled by 4×4 to the cache resolution of the teacher, which lowers the training-set SeK of the cached targets by 0.22 points on SECOND and by 2.7–3.4 points on the folds. (ii) *Ground truth only* (Landsat-SCD folds 1–3, seeds 1–3). The change target is the ground truth average-pooled to the cache resolution and upsampled, as in the screening controls. The semantic target is the label distribution smoothed across classes, $(1 - \varepsilon)\,\mathbf{e}_y + \varepsilon/(C - 1)$ on labelled changed pixels and uniform elsewhere, smoothed spatially in the same way, with ε chosen so that the true class receives the mean probability that the fold teacher assigns to it at $T_{\text{s}} = 2$ on the training set ($\varepsilon = 0.21$–0.28). This scalar is the only quantity taken from the teacher. We ran this control on the location-disjoint folds because they played no role in any

design choice. (iii) *Learning-rate schedule.* The undistilled student is retrained on SECOND with a cosine decay to 10^{-6} after the same warm-up (seeds 1–3).

V. Results on SECOND

A. Screening on the Validation Split

Table III and Fig. 4 summarise the screening. Change distillation alone raises the validation SeK with its weight up to $\lambda_{\mathrm{chg}} = 8$ (25.95 against 23.61 without distillation; the differences between 2, 4 and 8 are within two-seed noise) and plateaus beyond (-0.07 at $\lambda_{\mathrm{chg}} = 16$). The two ground-truth controls at $\lambda_{\mathrm{chg}} = 2$ give +0.04 and -0.02 points, against +1.83 for the teacher's targets at the same weight. For the change term, the gain is therefore not explained by adding a change loss of that weight or by softened boundaries; the controls were run only at $\lambda_{\mathrm{chg}} = 2$, with two seeds, on validation. Adding semantic distillation at $T_{\mathrm{s}} = 2$ brings a further +0.80 (all pixels) or +0.75 (changed pixels) points; $T_{\mathrm{s}} = 4$ is weaker (+0.45, +0.51). The two masks are equivalent within noise; the rule retained all pixels. Our feature-distillation term on top of output distillation adds +0.01, +0.12 and -0.45 points for the three variants, below the 0.36-point threshold, while multiplying the training time by 2.3–2.4 and the GPU memory by four; it was not retained.

B. Test Confirmation

Table IV and Fig. 5 report the test results. Without distillation, the student reaches 23.90 ± 0.15 SeK (best epoch, $n = 7$) and 23.65 at the last epoch. Change distillation alone gives 25.59 ± 0.13 (+1.69 [1.51, 1.88]). The full recipe gives 26.20 ± 0.11 (+2.30 [2.13, 2.47], Welch $p < 10^{-4}$), with Fscd 66.29 and mIoU 74.57. The semantic term contributes +0.61 [0.42, 0.80] points paired by seed ($p = 0.0009$, 5/5 seeds positive). The selection-free last-epoch number tells the same story: 26.20 against 23.65 (+2.55 [2.33, 2.77]). The distilled student's best and final SeK differ by at most 0.02 points (best epochs 108–119), whereas the undistilled student peaks between epochs 47 and 100 and then declines under the constant learning rate (Fig. 5b). Distillation thus raises the peak and removes the late decline; a decaying learning-rate schedule might remove part of that decline without distillation, which we did not test, so we base the gain on the best-epoch number.

Against the teacher (26.11), the distilled student differs by +0.09 [-0.04, +0.23] SeK; its 90 % interval [-0.01, +0.20] lies within ± 0.25 SeK, so the two are equivalent at that margin (two one-sided tests, $p = 0.02$), treating the single-run teacher as fixed. Against the teacher's selection-free last-epoch value (25.85), the student's last-epoch SeK (26.20) is higher. Its Fscd is slightly below the teacher's (66.41) and its mIoU slightly above (74.33). Because the distilled runs were still improving at epoch 120, we resumed them to 200 epochs, a decision taken after seeing the test curves and reported as a post-hoc extension: they reach 26.32 ± 0.11 (+0.22 [+0.08, +0.35] above the teacher, $p = 0.01$) and plateau after about 150 epochs, while the undistilled student's best epoch is unchanged and its final SeK drops to 23.30.

C. A Second, Smaller Teacher

With the same recipe and no re-tuning, the VMamba-B teacher (25.31 SeK, 113.01 M parameters, ImageNet-pretrained rather than a remote-sensing foundation model) yields a student at 25.69 ± 0.17, which exceeds that single-run teacher by +0.38 [+0.16, +0.59] ($p = 0.009$). The ViT-G teacher, more accurate and $33\times$ the student's size, yields the better student: +0.51 [0.27, 0.76] SeK paired by seed ($p = 0.004$, 5/5 seeds). The two teachers differ at once in size, architecture, pre-training and accuracy, so this comparison does not test the capacity-gap hypothesis [13], [14]; it shows only that a much larger teacher did not yield a worse student here.

D. Efficiency

Table V compares the two models on the same A100 at 512×512 (median of 50 runs after 10 warm-up runs). With the same operation counter on both models, the student needs $51.2\times$ fewer MACs (29.48 against 1509.74 GMACs; the counter omits the student's selective scan and the teacher's deformable attention, both small). The latency advantage is smaller: $12.5\times$ in float32 and $4.9\times$ in bfloat16 at batch size 8, $10.2\times$ at batch size 1 in float32. The teacher benefits from tensor cores in half precision, whereas the student is limited by memory accesses; latency, which governs deployment, does not follow the MAC count. Peak memory at batch size 8 is 1.96 against 25.09 GB.

E. Comparison with Published Results

Fig. 1 and Table VI place the distilled student among published SECOND results obtained on the official split. Published values are taken from the papers. ChangeMamba and Mamba-FCS report fvcore counts labelled "GFLOPs", which count multiply–accumulates, so we plot them as GMACs; conventions are not uniform across papers (PerASCD reports 2853 "GFLOPs" for the model we measure at 1509.74 GMACs), so cross-paper costs are indicative only. The distilled student reaches the accuracy level of the foundation-model teacher at the lowest compute in the table. Published accuracies are single numbers whose selection rules are not always stated.

VI. Results on Location-Disjoint Landsat-SCD

Table VII and Fig. 6 report the four folds.

Teachers. The retrained fold teachers pass the gate on folds 1–3 and fail it on fold 0, where the teacher's validation SeK (21.85) is below that of the undistilled students (22.50). On the test tiles, the teachers lead the undistilled students by +0.45, +1.00 and +1.75 SeK on folds 1–3 and trail by -0.14 on fold 0, a mean lead of 0.77 points, far below the 2.2-point lead on SECOND. Because the rule fixed before training the teachers caches only teachers that pass the gate, folds 1–3 form the primary analysis. We also ran fold 0 with the gate explicitly lifted and marked as exploratory, and report it as a secondary analysis.

Distillation. Distillation improves the student on every fold and every seed. On the three primary folds, the test SeK rises

TABLE III
SCREENING ON THE SECOND INTERNAL VALIDATION SPLIT.

Arm	λ_{chg}	Semantic KL	Feature KD	SeK max	SeK final	Δ max vs no KD (p)	h/run	GB
no KD	0	–	–	23.61 ± 0.30	23.25 ± 0.26	–	4.8	4.2
control: binary GT target	2	–	–	23.65 ± 0.07	22.78 ± 0.14	+0.04 (0.89)	4.7	4.2
control: smoothed GT target	2	–	–	23.59 ± 0.01	23.06 ± 0.69	−0.02 (0.94)	4.7	4.2
change KD	0.5	–	–	24.20 ± 0.54	24.01 ± 0.27	+0.59 (0.04)	4.8	4.2
change KD	1	–	–	24.86 ± 0.46	24.66 ± 0.17	+1.25 (<0.001)	4.9	4.2
change KD	2	–	–	25.44 ± 0.21	25.31 ± 0.03	+1.83 (<0.001)	4.7	4.2
change KD	4	–	–	25.53 ± 0.34	25.49 ± 0.30	+1.92 (<0.001)	4.7	4.2
change KD	8	–	–	25.95 ± 0.40	25.79 ± 0.44	+2.33 (<0.001)	4.7	4.2
change KD	16	–	–	25.88 ± 0.01	25.82 ± 0.03	+2.26 (<0.001)	4.7	4.2
DistilSCD (output KD)	8	T = 2, all	–	26.74 ± 0.06	26.66 ± 0.13	+3.13 (<0.001)	4.7	4.2
output KD	8	T = 2, changed	–	26.70 ± 0.12	26.69 ± 0.13	+3.09 (<0.001)	4.7	4.2
output KD	8	T = 4, all	–	26.40 ± 0.18	26.28 ± 0.02	+2.79 (<0.001)	4.7	4.2
output KD	8	T = 4, changed	–	26.46 ± 0.31	26.45 ± 0.32	+2.85 (<0.001)	4.7	4.2
+ feature KD	8	T = 2, all	1/8, 1/16; 1	26.76 ± 0.15	26.67 ± 0.25	+3.15 (<0.001)	11.2	17.4
+ feature KD	8	T = 2, all	1/8, 1/16; 4	26.29 ± 0.14	26.29 ± 0.14	+2.68 (<0.001)	10.8	17.4
+ feature KD	8	T = 2, all	1/8, 1/16, 1/32; 1	26.86 ± 0.20	26.82 ± 0.13	+3.25 (<0.001)	10.7	17.4

16 arms × 2 seeds (1, 2) = 32 runs, 120 epochs, trained on 2 671 SECOND training pairs and evaluated on the 297-pair internal validation split (the test set was not used). Mean ± standard deviation. Δ: difference of the mean max SeK to the no-KD arm, with the p-value of a pooled-variance t-test. Controls replace the teacher's change target by the binary or smoothed ground truth, same loss and weight. Feature KD: encoder stages (fraction of the input resolution) and weight λ_{feat}, on top of the retained output KD.

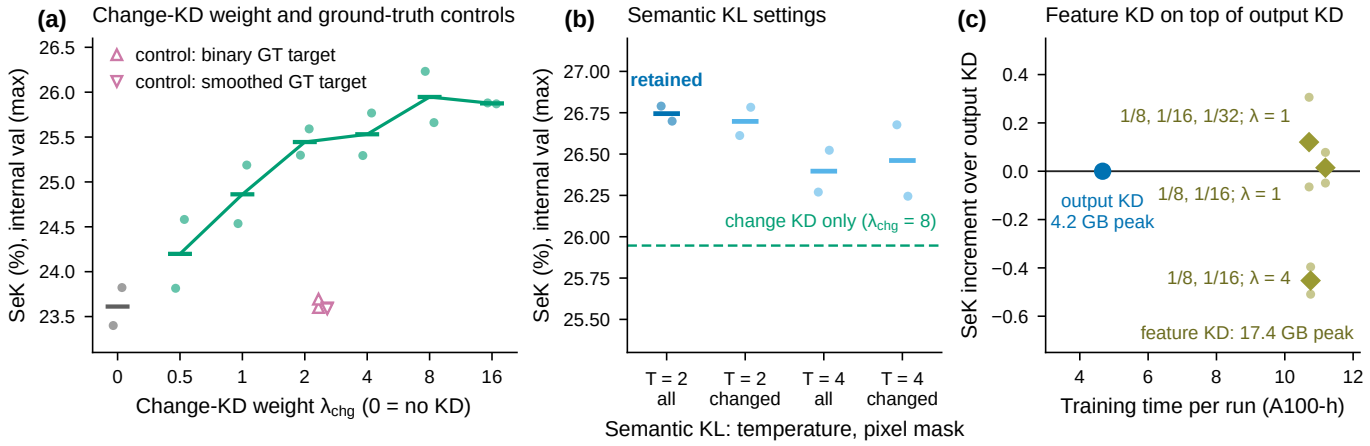


Fig. 4. Screening on the SECOND internal validation split (2 671/297 pairs; the test set was not used); points are seeds (two per arm), ticks are means, y is the best validation SeK over 120 epochs. (a) Weight of the change term; the two controls at $\lambda_{\text{chg}} = 2$ replace the teacher's target by the binary or the smoothed ground truth. (b) Semantic KL on top of change distillation ($\lambda_{\text{chg}} = 8$, dashed line): temperature and pixel mask. (c) Feature distillation on top of the retained output distillation: increment paired by seed versus training time per run.

TABLE IV
RESULTS ON THE SECOND TEST SET.

Model	n	SeK max	SeK final	Fscd	mIoU	Gain [95 % CI]	− teacher [95 % CI]	p
Student, no KD	7	23.90 ± 0.15	23.65 ± 0.22	64.42 ± 0.16	73.23 ± 0.08	–	–	–
Change KD only (ViT-G)	5	25.59 ± 0.13	25.57 ± 0.13	65.62 ± 0.14	74.37 ± 0.06	+1.69 [1.51, 1.88]	−0.52 [−0.68, −0.35]	<0.001
DistilSCD (ViT-G)	5	26.20 ± 0.11	26.20 ± 0.11	66.29 ± 0.11	74.57 ± 0.05	+2.30 [2.13, 2.47]	+0.09 [−0.04, 0.23]	0.13
DistilSCD (VMamba-B)	5	25.69 ± 0.17	25.69 ± 0.18	65.73 ± 0.14	74.38 ± 0.09	+1.79 [1.57, 2.01]	+0.38 [0.16, 0.59]	0.009
Teacher ViT-G (548 M)	1	26.11	–	66.41	74.33	–	–	–
Teacher VMamba-B (113 M)	1	25.31	–	65.61	74.01	–	–	–

Mean ± standard deviation over seeds, 120 epochs. SeK max: maximum test SeK over epochs (field convention, as in ChangeMamba and PerASCD); Fscd and mIoU at that epoch. SeK final: last epoch, no selection. Gain: Welch difference of SeK max to the no-KD student. − teacher: one-sample difference of SeK max to the teacher of the arm (ViT-G for change KD), with its p-value. Final-epoch gains are given in the text. Teachers: published checkpoints re-evaluated under our protocol, single run. Post-hoc 200-epoch extension of DistilSCD (ViT-G): 26.32 ± 0.11 (Supplement).

by 2.53 points (95 % interval over fold means [0.74, 4.32], $p = 0.03$; 9/9 fold–seed pairs positive), with Fscd +1.30 and mIoU +1.46. Over all four folds the gain is 2.51 [1.58, 3.45] ($p = 0.003$, 12/12 pairs positive). Per fold, the gains are 2.48, 1.84 and 3.28 on folds 1–3 and 2.46 on the exploratory fold 0, where the teacher was no better than the student. The spread across seeds is smaller with distillation on three of the four folds.

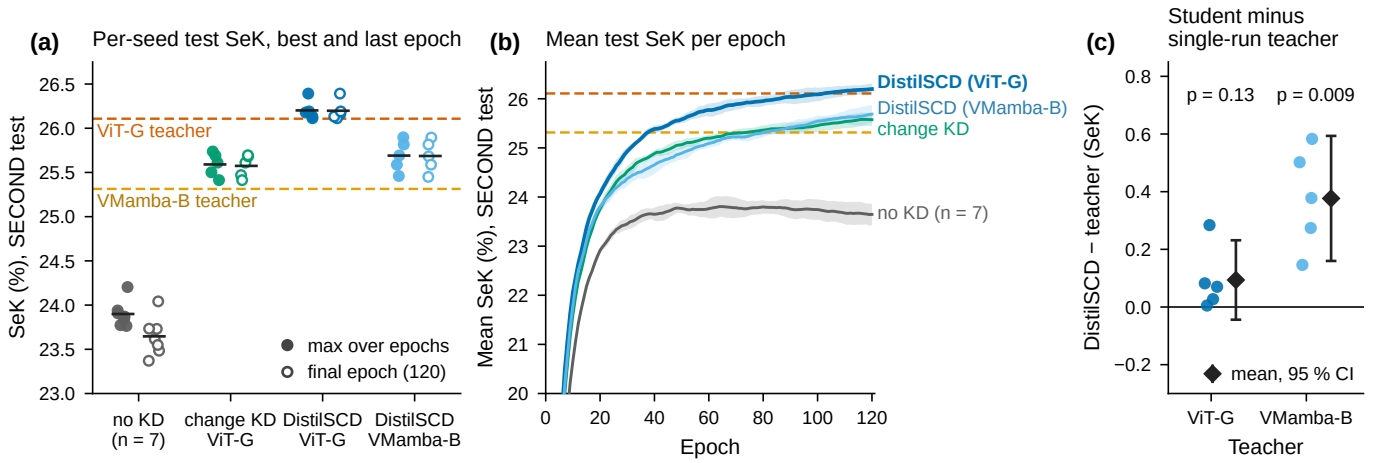


Fig. 5. SECOND test results. (a) Per-seed test SeK, best epoch (filled) and last epoch (open), for the student without distillation ($n = 7$), change-only distillation and DistilSCD with each teacher ($n = 5$); ticks are means; dashed lines are the single-run teachers. (b) Mean test SeK per epoch (band: ± 1 s.d. across seeds); the undistilled curves are the first 120 epochs of 200-epoch constant-rate runs. (c) DistilSCD minus teacher per seed, with mean and 95 % interval: +0.09 with ViT-G, +0.38 with VMamba-B; teachers are single runs.

TABLE V
EFFICIENCY OF DISTILSCD AND ITS VIT-G TEACHER ON ONE A100 GPU: (A) INFERENCE, (B) TRAINING COST PER RUN (120 EPOCHS).

(a) Inference

Quantity	DistilSCD	Teacher	Ratio
Parameters (M)	16.5	548.2	33.3×
GMACs / pair (aten counter)	29.48	1509.74	51.2×
GMACs / pair (fvcore)	31.42	–	–
Latency b8 fp32 (ms)	129.4	1613.4	12.5×
Latency b8 bf16 (ms)	113.7	561.1	4.9×
Latency b8 fp16 (ms)	112.6	487.1	4.3×
Latency b1 fp32 (ms)	26.3	268.5	10.2×
Latency b1 bf16 (ms)	30.2	125.6	4.2×
Peak memory b8 fp32 (GB)	1.96	25.09	12.8×
Peak memory b8 bf16 (GB)	1.82	26.09	14.3×

Same job and GPU for both models; 512×512 pairs; median latency; b8 / b1: batch of 8 / 1 pair(s). The teacher's fvcore count failed (JIT tracing of bicubic interpolation). Both counters ignore custom CUDA operators.

(b) Training

Prot.	Arm	Runs	h/run	s/epoch	GB
confirm.	change KD	5	6.38	149.3	4.23
confirm.	DistilSCD (output KD)	5	6.43	150.5	4.23
screen.	no KD	2	4.84	138.7	4.23
screen.	DistilSCD (output KD)	2	4.67	133.4	4.23
screen.	+ feat. KD 1/8, 1/16; 1	2	11.19	319.9	17.42
screen.	+ feat. KD 1/8–1/32; 1	2	10.71	316.3	17.43
screen.	+ feat. KD 1/8, 1/16; 4	2	10.76	317.4	17.42

Pairs used (confirm.: 2968 train, test set as validation; screen.: 2671 train, 297 val). GB: peak training memory. From the run timing logs. Output KD reads the offline logit cache and adds no measurable training time; feature KD runs the teacher online.

Distilled student versus teacher. The distilled student exceeds its fold teacher on all four folds, by 1.46 points on average over folds 1–3 ($p = 0.05$ with three folds) and 1.75 over all four ($p = 0.02$). Each fold teacher is a single run, trained with a recipe that its authors tuned on the random split and selected on a single validation tile, so these teachers may be under-tuned. The comparison does not depend on how the teachers' epoch was selected: the distilled students exceed even the teachers' best test epoch on every fold, by at least 0.22 points (supplementary Table SIV).

Budget. The 100-epoch budget was fixed on undistilled students; it does not handicap distillation: the validation SeK of the distilled runs changes by at most 0.10 points per 10 epochs over epochs 70–99, and the last-epoch test gain (2.48 on folds 1–3) equals the gain at the selected epoch.

VII. CONTROL EXPERIMENTS AND QUALITATIVE RESULTS

Table VIII reports the controls of Section IV-G.

Self-distillation. On SECOND, distilling the undistilled student into new seeds raises SeK by +0.75 [+0.41, +1.08] at the best epoch and by +1.05 at the last epoch, and the self-distilled students exceed their own teacher by +0.75 points, a born-again effect [15]. The PerASCD targets add +1.52 [+1.35, +1.70] SeK on top (3/3 seeds), so self-distillation accounts for about 33 % of the gain on the same seeds. On the location-disjoint folds, self-distillation gives no gain (−0.04 [−0.86, +0.77]; 1/6 fold–seed pairs positive), and the PerASCD targets exceed it by +2.61 [+1.73, +3.49] (6/6). Pooling degrades the student's cached targets more on Landsat-SCD than on SECOND (Section IV-G), which may contribute to this null result; the next control does not share that limitation.

Ground truth only. Targets built from the labels alone, softened spatially like the teacher's and across classes to the teacher's mean confidence, do not improve the student (−0.21 [−1.04, +0.61], 4/9 pairs positive; last epoch −0.30). With the same losses, weights and cache, the PerASCD targets exceed them by +2.75 [+1.55, +3.94] SeK, on 9/9 fold–seed pairs. This extends the screening controls, which concerned the change term only at $\lambda_{\text{chg}} = 2$ on validation data seen by the teacher, to both terms at the retained configuration, on data that played no role in its selection.

Learning-rate schedule. A cosine schedule lowers the undistilled student's SeK (−0.33 [−0.68, +0.03] at the best epoch, −0.82 [−1.08, −0.56] at the last), so the constant learning

TABLE VI
COMPARISON WITH PUBLISHED RESULTS ON THE SECOND TEST SET.

Model	Params (M)	GMACs	Counter	SeK	Fscd	mIoU	Provenance
PerASCD, ViT-G (teacher)	548.17	1509.74	aten	26.11	66.41	74.33	re-evaluated
Mamba-FCS	189.54	263.15	paper	25.50	65.78	74.07	reported, checked
VMamba-B + CG-Decoder (teacher)	113.01	352.90	aten	25.31	65.61	74.01	re-evaluated
MambaSCD-Tiny, released ckpt.[a]	37.13	115.44	fvcore[a]	23.34	–	–	measured
SCanNet (retrained by PerASCD)[b]	27.90	264.95	paper	23.18	63.53	73.03	reported, checked
MambaSCD-Base	89.99	211.55	paper	22.92	–[c]	73.68	reported, checked
MambaSCD-Tiny	21.51	73.42	paper	22.08	–[c]	73.33	reported, checked
Student, no KD	16.48	29.48	aten	23.90	64.42	73.23	ours, 7 seeds
DistilSCD (ViT-G teacher)	16.48	29.48	aten	26.20	66.29	74.57	ours, 5 seeds
DistilSCD (VMamba-B teacher)	16.48	29.48	aten	25.69	65.73	74.38	ours, 5 seeds

Params in M; GMACs per image pair at 512×512. aten: torch.utils.flop_counter, the same counter for our models and the teachers (measured by us); it does not count the deformable attention of the teacher nor the selective scan of the student (fvcore gives 31.42 GMACs for the student). paper: complexity reported by the original papers as 'GFLOPs', which are fvcore-style MAC counts for ChangeMamba and Mamba-FCS (conventions differ across papers); a different counter from aten, so not strictly comparable. Accuracies of published models as reported in their papers, on the official SECOND split. [a]Released checkpoint evaluated with our metric code; params and fvcore GMACs measured by us; the released code builds a larger configuration than the one reported in the paper (row MambaSCD-Tiny). [c]ChangeMamba reports an 'F1' column whose values are not comparable with the Fscd of the other rows; omitted. [b]SCanNet retrained on the official split by the PerASCD authors (their Table I; params and GFLOPs from ChangeMamba Table VIII); the SeK reported in the SCanNet paper uses a different, non-official evaluation split and is not used. Provenance: reported, checked = from the cited paper, checked against it re-evaluated = published checkpoint evaluated under our protocol (SeK, Fscd, mIoU as published, SeK reproduced); measured = released checkpoint run with our code. Sources: PerASCD (arXiv 2602.13780, Table I), Mamba-FCS (arXiv 2508.08232, Tables V–VI), ChangeMamba (arXiv 2404.03425, Tables VII–VIII).

TABLE VII
LANDSAT-SCD, LOCATION-DISJOINT FOLDS: TEST SEK.

Fold	Test tiles	Gate	Teacher	Student, no KD	DistilSCD	Gain	KD − teacher	Teacher − student
0[†]	00, 02, 20	no	28.61	28.75 ± 0.37	$\mathbf{31.21 \pm 0.09}$	+2.46	+2.60	−0.14
1	01, 21, 30	yes	26.02	25.57 ± 0.08	$\mathbf{28.05 \pm 0.15}$	+2.48	+2.03	+0.45
2	10, 12, 22	yes	22.48	21.48 ± 0.32	$\mathbf{23.32 \pm 0.08}$	+1.84	+0.84	+1.00
3	11, 31, 32	yes	20.18	18.42 ± 0.60	$\mathbf{21.70 \pm 0.17}$	+3.28	+1.52	+1.75
Folds 1–3 (primary)			22.89	21.82	24.36	+2.53 [0.74, 4.32]	+1.46	+1.07
Folds 0–3 (secondary)			24.32	23.56	26.07	+2.51 [1.58, 3.45]	+1.75	+0.77

Test SeK at the epoch of best validation SeK of the fold. Students: mean ± standard deviation over 3 seeds; teacher: PerASCD ViT-G retrained by us with the authors' recipe, one run per fold. Gain: KD − no KD, paired by seed; fold means for the summary rows, with a 95 % CI across folds. Positive seed pairs: 9/9 (folds 1–3), 12/12 (folds 0–3; a consistency count, seeds within a fold are not independent). [†]Fold 0 failed the teacher gate specified in advance and was run with the gate lifted (exploratory).

rate of the recipe does not handicap the baseline; DistilSCD exceeds the cosine baseline by +2.70 SeK.

Qualitative results. Fig. 7 shows test pairs chosen by a fixed rule, without visual selection. On SECOND, the distilled maps are less fragmented than those of the undistilled student and recover changed regions that it misses (fifth pair: water at T1), while some errors of the undistilled student persist (sixth pair: land cover at T1, which the teacher gets right). On Landsat-SCD, the maps of all three models remain fragmented, in line with the low test SeK under the location-disjoint protocol.

VIII. DISCUSSION

A. Gain and Teacher Advantage

One might expect distillation to help in proportion to how much better the teacher is than the student. Within SECOND, the more accurate teacher (advantage 2.21 against 1.41 SeK) does give the larger gain (+0.51 paired). Across our six teacher–dataset configurations (Fig. 6b), however, the gain stays between 1.8 and 3.3 SeK while the teacher's advantage ranges from −0.14 to 2.21, and on fold 0 a teacher slightly worse than the student on the test tiles still yields +2.46 points. Six configurations from two datasets with different selection rules cannot establish that the gain is independent of the teacher's advantage (correlation $r = 0.09$, $n = 6$); they do show that a teacher need not outscore the student to help it. In hindsight, our gate ("the teacher must beat the student on validation") was conservative.

What, then, is transferred? The controls of Section VII narrow the answer. On the location-disjoint folds, neither the student's own outputs nor labels softened to the teacher's confidence help, whereas the foundation-model targets add more than two SeK over both, so the benefit is tied to the teacher's predictions rather than to the form of the loss, the softness of the targets or the cache. On SECOND, about a third of the gain is obtained without a stronger model, as in born-again networks and the regularisation account of distillation [15], [16], [18]; the rest requires the foundation model. A teacher of the student's own accuracy (self-distillation: 23.87 against 23.90 SeK on SECOND) thus yields much less than PerASCD, whereas on the exploratory fold 0 a PerASCD teacher no better than the student on the test tiles yields a gain of the same size (+2.46). What matters is therefore less the teacher's score than information in its predictions that the student does not obtain from the labels. One candidate is the semantic term on the 80 % of SECOND pixels that carry no label; the teacher's targets are not a copy of the labels (it scores 33.24 SeK on its own training set). Which parts of the predictions carry the benefit remains to be tested.

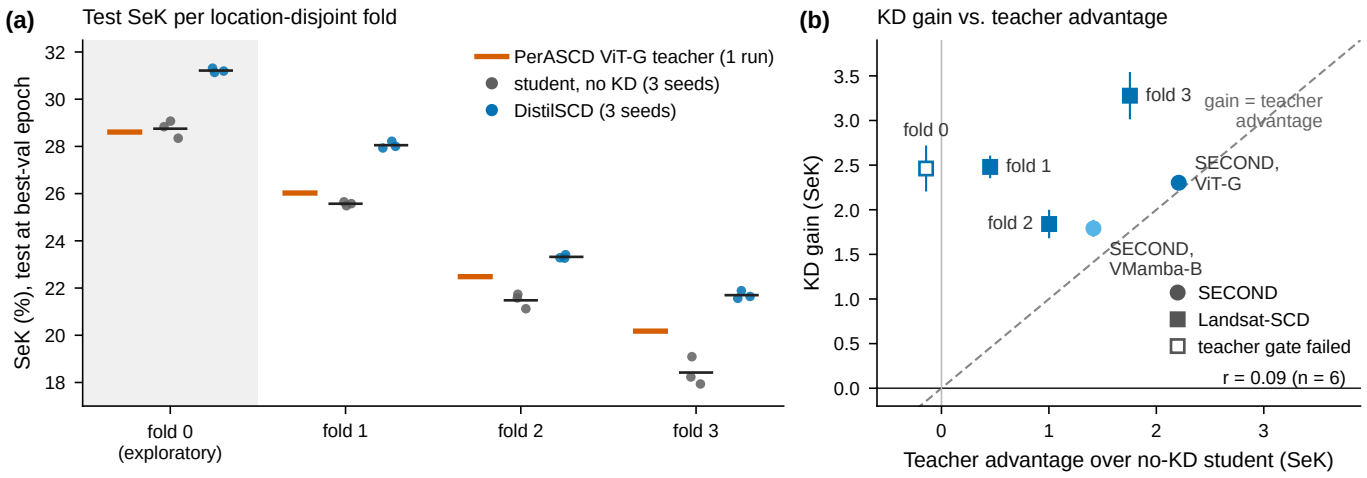


Fig. 6. Landsat-SCD, location-disjoint folds. (a) Test SeK at the epoch of best validation SeK: fold teacher (one run), student without distillation and DistilSCD (three seeds each). Fold 0 (shaded) failed the teacher gate specified in advance and is exploratory. (b) Distillation gain versus the teacher's advantage over the undistilled student, for six teacher–dataset configurations (SECOND: best test epoch; Landsat-SCD: three seed pairs per fold); error bars: ± 1 s.e.; open marker: fold 0; points above the dashed line are students that outscore their teacher.

TABLE VIII
CONTROL EXPERIMENTS: SOURCE OF THE DISTILLATION TARGETS AND LEARNING-RATE SCHEDULE (SEK).

Arm	Targets (teacher SeK)	Seeds	SeK	Last ep.	Δ vs no KD	Δ, last ep.	DistilSCD − arm
SECOND test, best test epoch (last epoch); paired by seed							
No KD, constant LR (recipe)	–	1–6, 42	23.90 ± 0.15	23.65	–	–	+2.35 [2.17, 2.53]
No KD, cosine LR	–	1–3	23.52 ± 0.12	22.79	−0.33 [−0.68, 0.03]	−0.82 [−1.08, −0.56]	+2.70 [2.65, 2.76]
Self-distillation	no-KD student (23.87)	2–4	24.62 ± 0.09	24.60	+0.75 [0.41, 1.08]	+1.05 [0.86, 1.23]	+1.52 [1.35, 1.70]
DistilSCD	PerASCD ViT-G (26.11)	1–5	$\mathbf{26.20 \pm 0.11}$	**26.20**	+2.35 [2.17, 2.53]	+2.65 [2.52, 2.78]	–
Landsat-SCD, location-disjoint folds 1–3, test at best validation epoch (last epoch); mean over folds, 95 % CI across fold means							
No KD	–	1–3	21.82	21.91	–	–	+2.53 [0.74, 4.32]
Ground truth only	none	1–3	21.61	21.61	−0.21 [−1.04, 0.61]	−0.30 [−0.83, 0.22]	+2.75 [1.55, 3.94]
Self-distillation	no-KD student	2, 3	21.77	21.80	−0.04 [−0.86, 0.77]	−0.10 [−0.99, 0.78]	+2.61 [1.73, 3.49]
DistilSCD	PerASCD, per fold	1–3	**24.36**	**24.39**	+2.53 [0.74, 4.32]	+2.48 [0.97, 3.99]	–

All arms use the same student, recipe, data pipeline and, for the distillation arms, the same losses ($\lambda_{\text{chg}} = 8$, $\lambda_{\text{sem}} = 1$, $T_{\text{s}} = 2$, all pixels) and the same eight-view cache format; only the source of the targets changes. Self-distillation: the teacher is the undistilled student of seed 1 (best test epoch on SECOND, best validation epoch on each fold); its students use other seeds. Ground truth only: change target = ground truth average-pooled to the cache resolution and upsampled; semantic target = label smoothing with ε matched to the teacher's mean confidence in the true class (0.21–0.28 across folds), uniform on unlabelled pixels. Δ: arm − no KD (constant LR), paired by seed (SECOND) or by fold and seed (Landsat-SCD); on SECOND, DistilSCD − no KD is paired over the five common seeds (unpaired over all seeds: Table IV).

B. Output Distillation and Our Feature Term

Feature distillation is often reported to help dense prediction [65], [67], and BFD [11] distils foundation-model features into binary change detectors. In our setting, once outputs are distilled, our feature term (cosine alignment of projected student features with the ViT-G encoder features) adds nothing at the tested weights and scales, and a larger weight hurts. We did not test feature distillation alone, other feature losses, or the generic foundation-model features that BFD distils. Output distillation also has a practical advantage: it can be computed offline, so the student's training cost is unchanged, whereas feature distillation needs the teacher online (2.3× the training time and four times the memory here).

C. What the Landsat-SCD Analysis Implies

Landsat-SCD numbers computed on random pair splits, ours included, answer a recognition question over a single, already-seen area. They remain meaningful for that question, but they should not be read as generalisation to new areas: on the released split, a lookup table exceeds the published models we compared. Under the location-disjoint protocol, scores drop to 18–31 SeK depending on the fold, the fold-to-fold variation is large, and conclusions require paired designs. We release the folds and the verification script so that future work can report both numbers.

D. Limitations

Our evidence covers two datasets, one student architecture and two teachers on SECOND that differ in several respects at once. Teachers are single runs: the published SECOND checkpoints, selected by their authors on the test set, and one retrained teacher per Landsat-SCD fold, trained with a recipe tuned by its authors on the random split. SECOND has no official validation split, so its headline numbers follow the field's test-epoch convention; the selection-free last-epoch numbers lead to the same conclusions. The student recipe was developed on SECOND, with test-epoch selection, before this study and then frozen; it uses a constant learning rate, and a cosine schedule did not improve the undistilled student. The screening used two seeds per arm and the controls two or three;

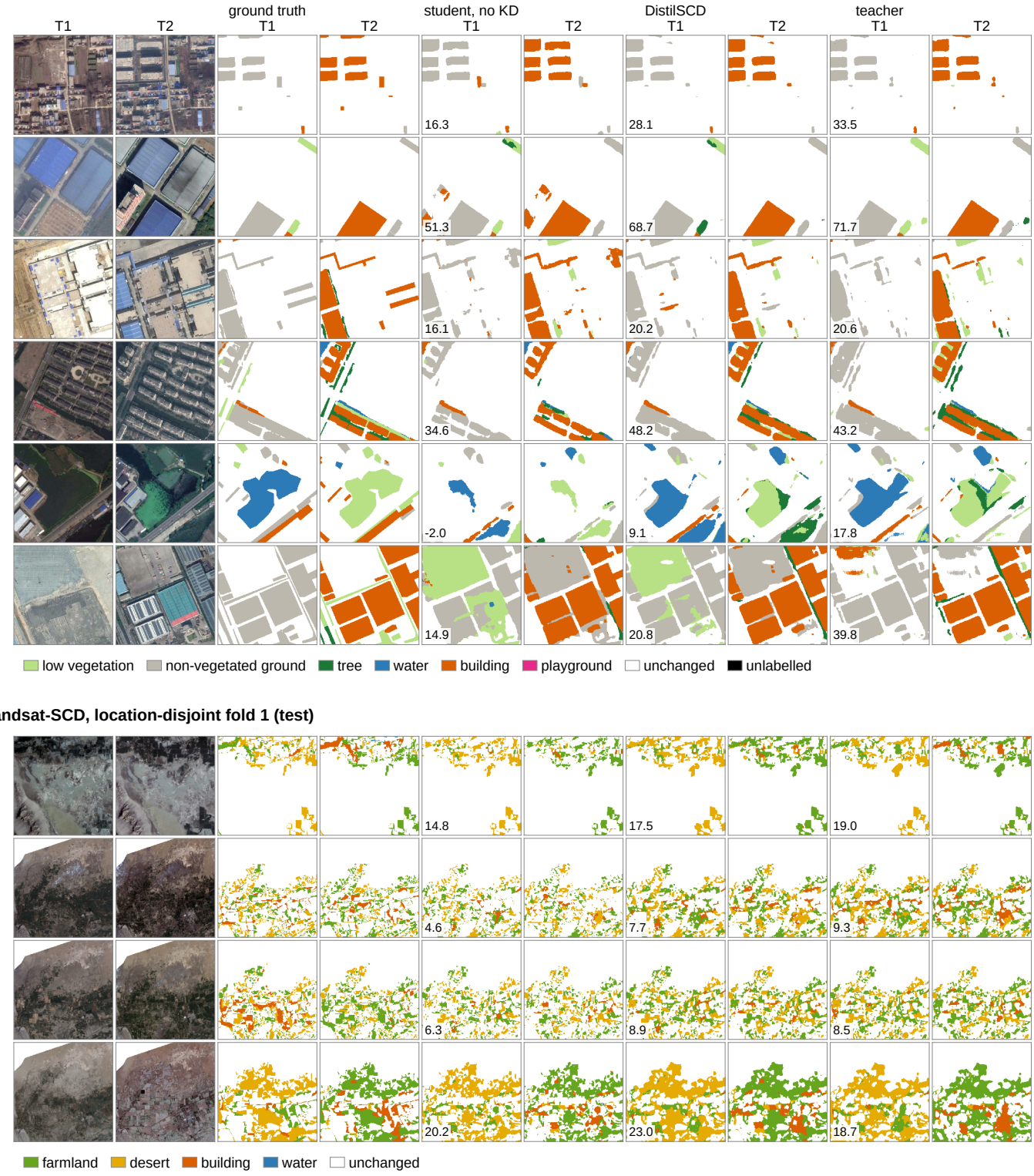


Fig. 7. Semantic change maps on test pairs: SECOND (top) and Landsat-SCD, location-disjoint fold 1 (bottom). Candidates are test pairs at evenly spaced quantiles of the changed fraction (24 on SECOND, 12 on fold 1); the rows are an evenly spaced subset of them, with no visual selection. Columns: T1 and T2 images, then the T1 and T2 maps of the ground truth, the undistilled student, DistilSCD (both seed 1) and the teacher (PerASCD ViT-G; on Landsat-SCD, the fold-1 teacher). White: unchanged; numbers: SeK of the pair (%).

the ground-truth-only control was run on Landsat-SCD only, and the self-distilled targets lost more accuracy to pooling on Landsat-SCD than on SECOND. Other distillation losses, other forms of self-distillation and a single-view cache were not tested. Each Landsat-SCD fold validates on a single tile, which makes model selection noisy, and the fold-0 analysis is exploratory. On Landsat-SCD, distillation first required training one 548.17 M-parameter teacher per fold (25.8 A100-hours in total). Finally, latency gains (4.2–12.5× on an A100) are much smaller than compute gains (51.2×), because the student is limited by memory accesses.

IX. Conclusion

Distilling only the outputs of a 548.17 M-parameter remote-sensing foundation model into a 16.48 M-parameter student brings the student to the teacher's accuracy on SECOND (equivalent within ± 0.25 SeK) at 51.2× fewer MACs, without changing the student or its inference cost. On Landsat-SCD, evaluating this required a location-disjoint protocol, because the released random split places every test image in the training set; under that protocol, distillation improves the student on every fold and seed and the distilled students exceed their single-run fold teachers. Controls with the same losses and cache tie the gain to the foundation model's targets: on the held-out folds, self-distillation and smoothed ground-truth targets yield none of it, and on SECOND self-distillation yields about a third. Future work should identify which parts of the teacher's predictions carry the benefit, test other student families, and extend location-disjoint evaluation to other change detection benchmarks.

Data and Code Availability

Code, training scripts, the SECOND internal validation split, the Landsat-SCD location-disjoint folds, the overlap-verification script and the per-run logs are available at https://github.com/mathismagron/distilscd. Student weights will be released once the licence of the teacher weights has been clarified with their authors. The teacher checkpoints are those released by the PerASCD authors; we do not redistribute them or the teachers we retrained from them.

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